



## Assessment Brief

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| <b>Module title:</b>             | Maths for Data Science                                                                                          |
| <b>Module code:</b>              | COM7023                                                                                                         |
| <b>Assignment title:</b>         | Maths for Data Science Portfolio                                                                                |
| <b>Assignment format:</b>        | Report                                                                                                          |
| <b>Word/time limit:</b>          | 3000                                                                                                            |
| <b>File type</b>                 | .docx or PDF file                                                                                               |
| <b>Percentage of final grade</b> | This assignment is worth 100% of your final grade for this module.                                              |
| <b>Submission deadline</b>       | See module iLearn page for date of submission                                                                   |
| <b>Grade release</b>             | You will normally receive your provisional grade and feedback within 20 working days of the submission deadline |

### Useful terms:

|                                |                                                                        |
|--------------------------------|------------------------------------------------------------------------|
| <b>Learning outcomes (LOs)</b> | The skills and knowledge that you should be able to show in your work. |
| <b>Rubric/Marking Matrix</b>   | A set of rules or guidelines used to grade or assess work.             |

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**Task summary:**

This assessment requires the submission of a written Mathematics for Data Science assignment. It evaluates your advanced understanding and application of key mathematical concepts underpinning data science, including Linear Algebra, Calculus, Probability, and Statistics.

You must demonstrate both theoretical knowledge and practical application within a data science context. All tasks should be presented in a single Word document, or PDF file, supported by screenshots of computational processes and results. Mathematical notation must be produced using appropriate software-based tools, and handwritten work is not permitted. All data analysis and visualisation must be carried out using Python, and your student ID must be clearly stated on the submission.

Failure to meet these requirements may result in penalties in line with programme regulations.

**Assignment instructions:**

You have recently joined **Urban Resource Intelligence (URI)**, a consultancy that advises local authorities on improving the **efficiency, sustainability, and resilience of urban resource systems**, including housing, utilities, and environmental services.

As part of your induction, your Line Manager has asked you to complete a mathematical investigation to demonstrate your ability to apply a range of advanced mathematical techniques to a real-world analytics problem before progressing to more complex data science projects.

The investigation is situated within the context of urban residential analytics, where city planners are seeking to better understand resource demand patterns, system stability, uncertainty, and behavioural differences across residential groups.

Your work will involve applying concepts from **Linear Algebra, Calculus, Probability, and Statistics** to analyse different aspects of the system and to communicate findings clearly to non-technical stakeholders involved in policy and planning decisions.

The initial stage of the investigation focuses on modelling steady-state resource demand across four interconnected residential zones within the city. The net daily demand (in standardised units) is modelled by the following system of equations:

$$2x_1 - x_2 + x_3 = 120$$

$$-x_1 + 3x_2 - x_4 = 150$$

$$x_1 + x_2 + 2x_3 - x_4 = 180$$

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$$-x_2 + x_3 + 2x_4 = 140$$

where  $x_1, x_2, x_3, x_4$  represent the average demand levels in each zone.

You are expected to analyse this system using appropriate linear algebra techniques, determine whether a stable equilibrium exists, and interpret the resulting distribution in the context of urban planning and load balancing.

Once the steady-state behaviour has been investigated, your manager asks you to analyse how demand changes over time within a high-density residential area. The rate of demand is modelled as a function of time  $t$  (hours) by:

$$D(t) = 4t^3 - 18t^2 + 24t + 90$$

You should analyse this function to examine rates of change, identify periods of rapid increase or decrease, and determine any critical points. Your discussion should explain how such analysis supports peak-demand management and operational planning.

During data collection, it is recognised that short-term forecasting errors are unavoidable due to behavioural variability and environmental factors. These errors are assumed to follow a normal distribution with a mean of **0** and a standard deviation of **5** units.

You are required to analyse this distribution to evaluate the likelihood of large deviations from expected demand levels and to interpret how these probabilities inform risk management and contingency planning.

Finally, the investigation examines whether average daily resource consumption differs between residential building types, which is critical for designing targeted efficiency interventions.

Daily average consumption data has been collected from three residential categories:

- Apartments:

16.8, 17.2, 16.5, 17.9, 18.1, 17.4, 16.9, 17.6

- Terraced houses:

18.4, 18.9, 19.1, 18.7, 19.3, 18.6, 19.0, 18.8, 19.2, 18.5

- Detached houses:

20.1, 19.7, 20.4, 19.9, 20.6, 20.2

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You should apply appropriate statistical techniques to determine whether mean consumption differs significantly between groups, clearly stating hypotheses, assumptions, and conclusions. Unequal sample sizes must be handled appropriately.

Once all numerical analyses have been completed, you must critically evaluate the mathematical approaches used across the investigation. Your final report should reflect on:

- Assumptions and limitations of the models
- Reliability of conclusions under uncertainty
- Practical implications for urban policy and planning

The completed report must be written for non-technical stakeholders within Urban Resource Intelligence and should focus on the meaning and implications of the results, rather than procedural detail.

**(100 Total marks)**

**(2000 words equivalent)**

**(1000 words for written results)**

**(LO's: 1,2,3 & 4)**

## Learning outcomes (LO)

By completing this assessment, you will have shown and be assessed on **all** four of the learning outcomes:

LO1. Develop an advanced understanding of mathematical concepts and their application in data science.

LO2. Apply mathematical principles to solve data science-related problems.

LO3. Deploy mathematical functions, models, and equations to derive advanced analytical solutions.

LO4. Identify and solve novel and complex problems related to aims and desired outcomes. Critically evaluate and reflect on the approaches and solutions identifying and embedding possibilities for originality or creativity.

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Your marker will use a rubric/ marking matrix to grade your work, and you can access this by clicking on the submission portal.

## Guidelines and policies

You can find links to more useful information about the assignment and university policies below.

### Word/time limit policy

[Click here to view the Arden University word count/time limit policy](#)

### Referencing guidelines

[Click here for Harvard referencing guidelines](#)

Please follow the referencing guidelines that are appropriate for your degree programme. If you are unsure which you should be using, please contact your module team.

### Academic integrity and misconduct policy

[Click here to view Arden University's policy on academic integrity and misconduct](#)

### Statement on use of artificial intelligence on assessment

[Click here to view Arden University's statement on the use of artificial intelligence on assessment](#)

### Support information

[Click here to view guidance on how to apply for short-term extensions](#)

[Click here to view guidance on how to apply for extenuating circumstances](#)

[Please click here for link to academic skills team support](#)